Response to Reviewers

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Young Children's Spontaneous Comprehension of Symbol-Referent Relationships in the Graphic Domain

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Manuscript ID: **CD MS CHILDDEV-2025-1273**

We are very thankful for the reviewers' comments and think the manuscript has highly benefited from their suggestions and questions. Below we provide a detailed overview of all points that were raised and how we responded. In order to make the review process more convenient, all points have been numbered (e.g. R2.4 might be read as *reviewer 2, point 4;* also see E.1. for points from the editor). All comments by reviewers and editors are in italics.

Editor's Notes

Overall Evaluation

All three reviewers brought up the theoretical clarity of the introduction, and would like to see revision. I agree. In its present form, the introduction does not present a clear, coherent description of the theoretical motivation for the present research. In some places, this simply involves being more precise with your choice of phrasing. In other places, however, this will involve significant conceptual reorganization and rewriting to ensure that you are motivating only the ideas that are necessary for the present study. I will encourage you to look carefully at the main points made by Reviewer 1, the major comment under the heading, "theoretical" by Reviewer 2, and points 1-3 by Reviewer 3.

E1.1

To offer you a concrete suggestion in your revising, the introduction is also a little long, and could be shortened by 500-1000 words. If you try to revise with that constraint in mind, it might help focus you on the specific phrasings and arguments to clarify your position.

Response:

Thanks! To address this we have reworked the introduction. Rather than going through the literature discussing all symbol-referent relationships tested across the 12 conditions of the three studies, we now start out with a very brief review of the development of communicative competences and then provide a detailed overview of the steps required in solving symbolic object-choice tasks. This allows us to introduce all relevant terminology and discuss the inferences children need to make when solving these tasks, separating pragmatic and analogical reasoning and reviewing results for both strands of literature separately. This in turn helps to highlight the contribution of our work. To focus only on ideas that are central to our investigation, we have suppressed references to children's production of drawings considerably shorten the discussion of stimulus dimensions.

E.1.2

Further, I want to remind you that not all readers will be familiar with the Bayesian analyses that you present. For the purposes of readability, you might want to step the reader through your analysis a little more clearly as well, to ensure that all readers are following the arguments you are making.

Response:

We have rewritten the paragraphs outlining the analyses. In addition to providing all information necessary for reproducing the analyses, we now also describe why the respective steps and decisions were made. While this may be a bit verbose for readers with expertise in Bayesian modelling, it should be very convenient for readers that are less familiar with this analytic approach:

> *To evaluate how children's ability to interpret cues varies by age and condition, we employed Bayesian Logistic Generalized Linear Mixed Models (GLMMs). This mode of analysis is appropriate as it allows to estimate the probability of success while accounting for uncertainty and individual differences, and because it allows mapping linear predictors to a probability between 0 and 1 via a link function. Our analyses modeled children's binary choices (correct vs. incorrect) using fixed effects for condition, age as well as their interaction as primary predictors. In addition, we included a fixed effect of trial number, to control for learning or fatigue, and a fixed effect of sex. To account for the possibility that some children may be better at the task than others, we included random intercepts for each participant. We also included random slopes for trial number, allowing potential effects of learning or fatigue to vary from child to child. To facilitate processing and interpretability, age and trial number were standardized to a natural mean of 0 prior to modeling. The full model notation was correct ~ condition*z.age + z.trial + sex + (z.trial | subid). In line with best-practice conventions, we evaluated the relevance of age and condition for children's performance prior to interpreting coefficients. For this, the full model outlined above was compared with a reduced model lacking the interaction of age and condition. To assess the fit and relative explanatory power of either model, we used Widely Applicable Information Criterion (WAIC) scores and weights (McElreath, 2018) as well as the difference in Expected Log Predictive Density (ELPD) via the function loo_compare (Sivula, Magnusson, Matamoros, & Vehtari, 2020). Across all studies, the preregistered full model proved superior or equally predictive. Model estimates were inspected for the different predictors including their 95% Credible Interval (CrI). In each study, the condition hypothesized as the most simple was set as the reference level within conditions. All models were implemented in Stan (Stan Development Team, n.d.) via the function brm of the package brms (Bürkner, 2017).*

E.2.1

Reviewer 2 brings up the absence of a control condition, which would better support the claim you are making. I want to highlight this, as a possible way you can expand on your *study by collecting additional data. If you want to collect such data, I would welcome it, and I suspect that would be a stronger way to revise. However, I will also accept changes in your argument structure and interpretation that acknowledge and address this concern in your revision.*

Response:
Due to changes in affiliations and the expiration of the project funding, we cannot collect additional data. Reviewer 2 is specifically asking for an additional study in which the same stimuli (or part of which) may be presented to children either within the framing of a symbolic object-choice task we employed ("The monkey will help you find the banana!") and contrastively in the manner of a standard match-to-sample task ("Look at these! Which ones go together?"). We have amended the manuscript by highlighting the closeness to match-to-sample tasks in the literature and again in the discussion as an avenue for future research in addition to a paragraph from the first draft already discussing the relations between our task and analogical reasoning.

E.3. General Requirements
Together with this cover letter providing a point-by-point reply and documentation of changes made for the revision, we submit

- a clean APA style manuscript
- a track changes version of the main document
- a lay summary (page 1 of the manuscript before the abstract)

SRCD sociocultural policies
The manuscript is compliant with the SRCD sociocultural policies. Under the *General Methods* section (p.10) we report (1) the dates of data collection; (2) race/ethnicity, socioeconomic status, language, (3) place(s) from which that sample was drawn, including country, region, and report the (4) selection and recruitment procedures.

StatCheck
Both manuscript and supplement have been run through StatCheck. The main manuscript does not contain significance tests. All exploratory t-tests reported in the supplements have been evaluated as "consistent" by StatCheck.

Exploratory vs. Confirmatory Declaration
In the General Methods section under Data Handling and Analyses (p.14ff) we outline that all analyses and hypotheses were preregistered. We highlight where our analyses deviate from the preregistered analysis (i.e. comparing models via ELPD differences). Exploratory analyses are clearly discussed as such and are presented only in the supplements. We highlight that we had hypotheses about the specific order of the age of success for the experimental conditions but did not make point prediction about the month of success.

Open Science Declaration.
All data, analysis scripts and materials are publicly available. This open science declaration will be included on the title page:

> *Scientific Integrity and Openness*: The data and code necessary to reproduce the analyses presented here are publicly accessible, as are the materials necessary to attempt to replicate the findings. Analyses were pre-registered. For data, analyses materials, and preregistrations see: [LINK MASKED FOR REVIEW]

Reviewer 1

Overall evaluation
The manuscript presents 3 studies that explore children's graphic symbol understanding and use in the age range of 3-7 years. It presents a novel approach with stimuli that are more abstract than previously used in this area of research. The use of more abstract symbol-referent relations heightens the works' relevance to analogical reasoning, a core cognitive component that is very likely a core component of pictorial symbol processing. This link has not been explored in the graphic symbol literature previously. I find this work to be of high importance; it addresses an important question about the process of graphic symbol understanding and use (the role of analogical symbol/referent relations), the stimuli and tasks are carefully designed to test the hypotheses, the statistical analyses are appropriate, and the interpretations are consistent with the results. Specific points for the authors to consider are found below.

R.1.1 Consistent use of the term referent
The term referent is sometimes used to mean the symbol. This may stem from the confusion that the symbol and referent in the studies are both pictures. If this seems confusing to the authors then perhaps consistently using the terms cue and target when referring to the stimuli used in these studies, and symbol and referent when referring to other studies investigating graphic symbols, would clear the confusion.

Response:
We have amended the manuscript in line with the suggestion and now use *symbol* and *referent* when discussing the literature, but *cue*, *target* and *distractor* when discussing our work. Thanks!

R.1.2 Clarify Conceptual Framework
The literature review covers a lot of terrain, however, the links between the points is not obvious, even in the summary on p. 9. Perhaps the authors could set up the overall conceptual framework that motivated the study at the beginning of the introduction, followed by the general aims, (what is novel and why is it an important question) then delve into the specific aspects that are as yet understudied in the context of graphic symbol development. In

my reading of the manuscript the unique contribution of the work is its relevance to the link between symbolic development and the cognitive skill of analogical reasoning, but this is not sufficiently developed in the introduction for that link to be obvious.

Response:
A systematic investigation of a comprehensive set of symbol-object-relationships tracking development continuously across the preschool years while using one coherent paradigm across several studies is our contribution to the literature. We are thankful for the suggestion to emphasize the link between symbolic development and analogical reasoning particularly. The introduction has been reworked and substantially shortened and instead explicitly treats the literature on relational and non-relational match-to-sample tasks as a point of reference for our contribution (see pages 8-9).

R.1.3 Cohesion of Introduction
In general, the flow of the points in the introduction needs to be improved. For example, on p. 4 the paragraph shifts from a review of studies where pictures are used to find objects (Callaghan, DeLoache work) to the point that pictures as symbols of pictures are rarely found. It is not clear why this is an interesting question.

Response:
Thank you, we have rewritten the introduction to improve its flow. This shift in topics has been deleted. After an overview reviewing the onset of symbolic competencies up to the third year, we first introduce the concepts and cognitive inferences relevant symbolic object-choice tasks. The literature on specific symbol-referent relationships is not reviewed stepwise but once in the context of symbolic communication and once in the context of match-to-sample designs.

R.1.4 Olson 1994
Olson (1994), The world on paper, is a seminal work on literacy and cognition which may be helpful for the authors to consider.

Response:
Thank you, this is very helpful. A central claim of Olson is that literacy provides access to new concepts and modes of thinking, with a strong focus on benefits for children's developing (meta-)analytic reasoning and linguistic awareness. We cite Olson (1994) now in the opening paragraph when referring to language and literacy acquisition as significant achievements in cultural learning (see page 3).

Olson, D. R. (1994). *The world on paper: The conceptual and cognitive implications of writing and reading*. Cambridge University Press.

R.1.5
Using the term 'realistic representation' is confusing – 'replica object' is typically used to denote these referents.

Response:
We replaced the phrase "realistic representation" in the discussion section with "*replica objects*". The introduction has been shortened considerably and the reference can now be found on page 5.

R.1.6
There are several points where clarification is needed about why a given statement is true. E.g., why are the perceptual dimensions of color, quantity, size etc easier to process than relational properties or orientation, position? Is this a perceptual mechanism at work? Or attentional? Is this mechanism operating at this age range?

Response:
Following the request to shorten the introduction by 500-1000 words, we have now omitted the statement entirely.

R.1.7
Delving more into the roots for your hypotheses – they do seem reasonable from classic perception and attention developmental work – is needed for the roots of the rationale to become clear. Some articles that may be informative along this line of reasoning are works that showed a shift from 3 to 5 to 7 years range in children's attention to whole objects vs. features of objects https://doi.org/10.1016/0022-0965(87)90057-9 *and research that showed interference in the processing of hue and form features early in the attentional process (pre-attention)* https://doi.org/10.1016/S0166-4115(08)60456-2 *. While not directly relevant as the current task is very different, these classic works may raise additional interpretations of the object/feature findings reported here. My understanding of the early work, in a nutshell, is that younger children process stimuli as wholes and only later are their attentional processes developed sufficiently to focus attention on specific features. This developmental trend, and interference when the task requires selective attention on a specific perceptual dimension of a stimulus (size, orientation, form, etc) may explain, for example, why the size of the feature was particularly difficult for children.*

Response:
Thank you, this is extremely helpful. We have incorporated these references in the interim discussions concluding the individual studies and refer to the shift from holistic to feature-oriented processing also in the discussion (pages 22, 26, 30, 34).

Callaghan, T. C. (1990). Texture segregation in young children. In *Advances in psychology* (Vol. 69, pp. 175-195). North-Holland.

Shepp, B. E., Barrett, S. E., & Kolbet, L. L. (1987). The development of selective attention: Holistic perception versus resource allocation. *Journal of Experimental Child Psychology*, *43*(2), 159-180.

Diamond, A., Carlson, S. M., & Beck, D. M. (2005). Preschool children's performance in task switching on the dimensional change card sort task: Separating the dimensions aids the ability to switch. *Developmental Neuropsychology*, *28*(2), 689–729. https://doi.org/10.1207/s15326942dn2802_7

R.1.8

The delineation of hypotheses was helpful, but it would also help to understand why these predictions were made. Reworking the introduction may make the rationale for the hypotheses more transparent.

Response:

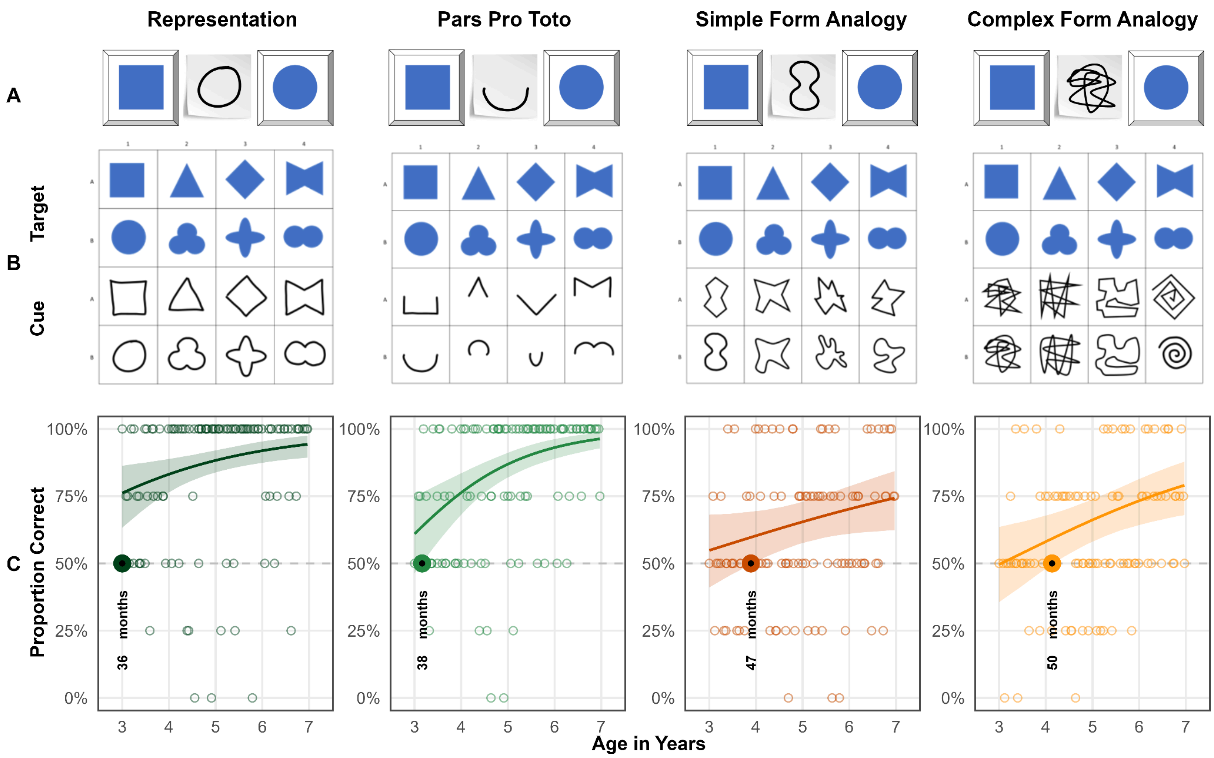
Thanks for pushing us to be more explicit here. We hope that the reworked introduction makes the motivation and basis for the hypotheses more clear. Our main aim, however, was to contribute to the literature by testing various symbol-referent relationships in one coherent paradigm and trace development through the preschool years. To be more explicit regarding the preregistered hypotheses but without adding to the word count of the introduction, we now provide short justifications of the hypotheses when stating them at the beginning of each study section on pages 17, 22 and 26ff..

R.1.9

For all figures it would be helpful to have the h/v axes labelled for B and C panels.

Response:

We have amended the illustrations accordingly, e.g. for study 1:



R.1.10

p. 6 – I did not understand the point in the last line of the paragraph beginning 'A highly abstract…' .

Response:

This is from a paragraph introducing analogies in form (round/edgy). The sentence has been removed and the introduction was reworked. Also and in line with suggestions from reviewer 3, the discussion of sound-symbolism has been shortened considerably. The paragraph now reads:

> *Another way of reducing iconicity may be the metaphorical use of form, such as using sharp or spiky drawings to refer to "pointy" concepts (Knoeferle, Li, Maggioni, & Spence, 2017). This phenomenon is central to research on sound symbolism within the lexicon (Bermúdez, 2020; Sidhu, 2025), as well as in studies of action and gesture (Bohn, Call, et al., 2019; Margiotoudi & Pulvermüller, 2020). Robust across cultures (Ćwiek et al., 2022), this sensitivity emerges around 30 months for word-shape matching (Fort et al., 2018; Imai, Kita, Nagumo, & Okada, 2008; Maurer, Pathman,*

& Mondloch, 2006) and by 36 months for matching actions with gestures (Bohn, Call, et al., 2019). Despite these findings, it remains unknown whether children can utilize the global properties or "impressions" of a shape's form to ground reference in the graphic domain alone.

R.1.11 *p. 7 – 'size words' is repeated*
Response: amended

R.1.12 *p. 9 – 48-60 months could be 4-5 years as years is the age unit throughout the paper.*
Response:
Thanks! We now use years wherever possible, except for when discussing changes occurring between 30 and 36 months when referring to the work by Judy Deloache and Tara Callagher, or when presenting our own findings.

R.1.13
p. 10 – it is misleading to indicate there are 96 children in the general methods as different numbers of children are tested in each of the studies. An important point of clarification is whether different children were tested in each study.

Response:
Thanks! During data collection, we have ensured that children can only participate once in any study contributing to this research project. Hence, different children were tested in each study. We had preregistered to collect a minimum of 96 children per study. We have amended the manuscript to make this more clear, for example, in the abstract:

> *Participants were <u>304</u> German-speaking children continuously sampled from age 3 to 7 (Study 1, N = 106, 51 female; Study 2, N = 99, 49 female; Study 3, N = 99, 55 female).*

as well as the general methods section:

> *In order to continuously trace the development of symbolic competences across the preschool years, for each study we aimed to test <u>a minimum</u> of two children per month of age between the third and the seventh birthday for a total of 96 participants balancing male and female participants.*

and when discussing the sample of study 2 and 3:

> *A total of 99 three- to seven-year-old children (M = 60.04 months, SD = 13.69 months, range 36 - 83 months; 49 female) participated [in study 2].* <u>*None of these had participated in study 1.*</u>

> *A total of 99 three- to seven-year-old children (M = 59.88 months, SD = 13.44 months, range 36 - 83 months; 55 female) participated [in study 3].* <u>*None of these had participated in the previous studies.*</u>

R.1.14
p. 14 – it is on this page that the use of referent became confusing.

Response:
Again, thanks for pointing out this lack of consistency. We now use *cue*, *target* and *distractor* throughout the manuscript.

R.1.15 *p. 15 – children's choices were logged by experimental script, or experimenter?*

Response: Children made their choice by touching either hiding place on screen. The experimental script immediately recorded which side they chose and choices were immediately logged as correct or incorrect according to the logic of the task. The manuscript was amended here to state this more explicitly:

> *Upon touching the screen, children's choices were logged and immediately coded as correct or incorrect by the code running the stimulus presentation.*

R.1.16
p. 18 – I wondered about the ages of children excluded – and see in the supplementary the breakdown of this. For Figure S1 – are excluded children included or excluded from this figure?

Response:
There is no overlap between the children in supplement figure 1 showing the ages of participants and supplement figure 2 showing the distribution of excluded children across the age-range. We have amended the manuscript by referring to the two supplement tables as indicating children included and excluded respectively.

R.1.17
Reference is made to the 'reference condition' throughout the results sections – it would help to specify the comparison condition as there are many conditions to track throughout the paper.

Response:

Yes, thanks! The general methods section states that the hypothetically most simple condition in each study was used as the reference condition in each model (study 1: representation, study 2: absolute position, study 3: size of object). When discussing the specific results we now suppress the ambiguous term "reference condition" but name the respective conditions explicitly.

R.1.18
p. 21 – In reference to the Simple and Complex Form Analogy stimuli, saying they are based on visual similarity seems to be a stretch. For example, sharing curved lines may link the drawing of a flower and a VW Beetle, but they would be judged as visually dissimilar.

Response: This is a very interesting point. Our experience from presenting this work in professional contexts is that researchers have very different intuitions about what iconicity or visual similarity are. While - as expressed here - some may be hesitant to refer to an "analogy in form" as visual similarity, others have discussed all stimuli we used as "being merely iconic", even including size and number .

To amend the paragraph we avoid this phrasing now and have replaced the sentence with:

> *While they are still based in iconicity, in so far as their interpretation involves a sense of resemblance with regard to their shape (Winter, Woodin & Perlmann, 2026), they are conceptually demanding at the same time.*

This is drawing on a recent contribution to the Oxford Handbook of Iconicity by Winter and colleagues (2026) suggesting a highly inclusive definition for iconicity in the cognitive sciences like "*A signal in any modality or medium exhibits iconicity when its production and/or interpretation involves a sense of resemblance between at least some aspect of its form and at least some aspect of its meaning.*" Given the more round or edgy shapes used in the form analogy conditions, there is at least a sense of resemblance to some aspects. Of course, there is a continuum from highly iconic to more conceptual symbol-referent relationships that we wanted to refer to when discussing the results.

R.1.19
p. 24 – My sense is that the condition Orientation of Feature would be better labelled Position of Feature as it has a top/bottom options and not directionality.

Response:
We have specifically opted for using only top/bottom options here as data from another study that we are currently preparing for publication is suggesting that children do indeed read such displays with a salient feature (a shape with a bump, dent, opening on one side) as being directional. Against this background, we would like to take the liberty of not changing the manuscript here. In advance, thanks for your understanding.

R.1.20
p. 25 – 'assign or abstract' – would 'infer' be a more apt term?

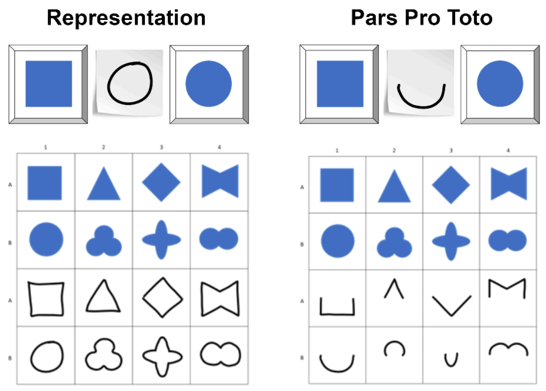
Response: The phrase has been replaced in line with the suggestion of the reviewer:

> *In order to correctly interpret the symbol-referent relationships employed here, children need to infer conceptual order in graphical displays.*

R.1.22
p. 31 – I do not understand the phrasing 'While cues are highly similar and targets even identical..'

Response:
Sorry for the confusion! The sentence compares the stimuli used in *Representation* and *Pars Pro Toto*. The targets (upper two rows in blue) are identical in both conditions and the cues (bottom two rows, outline drawings) are cut in half. The amended paragraph now reads:

> A second condition, *Pars Pro Toto*, used the same stimuli as *Representation*, but the cues provided only half of the respective figures to guide children's choices (cf. Figure 2 B, boxes 1 & 2). Despite using the same target stimuli as *Representation* and very similar cues, children succeeded in *Pars Pro Toto* no sooner than at 38 months of age.



R.1.23
p. 33 – the link between the trajectory of EF skills development and the current findings needs to be fleshed out more, It is difficult to see the link as it is phrased. In general ,these points are very interesting interpretations – especially reference to analogical reasoning work, but without more framing in the Introduction readers will need to take these statements at face value.

Response:
We have now added references with examples for how EF skills and communicative and analogical reasoning interact. We agree that being more precise here adds to the manuscript but would like to keep it as brief as possible to comply with requests for shortening the manuscript, but also since the importance of EF skills in match-to-sample, reading, drawing and decoding graphic displays is well documented.

In the introduction we highlight the role of EF skills in match-to-sample tasks and added references (see pages 8-9). In the discussion, we outline how components of EF may help with the demands of our task:

> *Specifically helpful for the tasks employed here may be developmental gains typically occurring around the fourth birthday in the domain of analogical reasoning (Christie & Gentner, 2014; Gentner, 1977; Shivaram et al., 2023), which are again fostered by a suite of executive function skills improving significantly during this important period in child development (Bohn, Tessler, Kordt, Hausmann, & Frank, 2023; Thibaut, French, & Vezneva, 2010). Focusing on various aspects of the stimulus-combinations, working memory (Richland et al., 2006; Simms, Frausel, & Richland, 2018) is crucial for keeping track of features children have explored. To succeed they are furthermore required to suppress irrelevant aspects of the stimuli and inhibit distracting impulses solving relatively abstract tasks (Carlson & Wang, 2007; Jablonski, 2014). Finally, cognitive flexibility is not only key for viewing the same stimulus from different conceptual angles but also specifically for switching modes of interpretation across conditions (Cuevas et al., 2014; Willoughby, Wirth, & Blair, 2012).*

R.1.24
p. 34 – Do the authors think that when you use highly abstract pictorial symbols in these tasks that the more abstract relations will pop out more easily than if realistic (or highly familiar shapes) pictorial symbols are used? It may be that we use basic filters as a first line of processing with pictorial symbols (color, form) because it works for most symbolic matching tasks, but pull out other filters (orientation, number, size) when the match uses more abstract symbols. The first paragraph on this page needs to be fleshed out – very interesting suggestions that raise more interesting possibilities.

Response:
Our hunch is that using realistic target stimuli of familiar objects will make it harder for children to find abstract relations. One reason for this may be that depictions of natural objects are necessarily more complex than simple geometric shapes. In addition, natural objects will trigger semantic processes (context, affordances, use cases). In both cases, cognitive load increases and may tax children's performance. An interesting case in question

that we are citing is Long et al.'s (2019) stroop-effects when asking preschoolers to make size comparisons with depictions of real-world objects that are congruent or incongruent with their actual size. The paragraph in question has been amended using the references suggested above (Shepp, Barrett & Kolbet, 1987; Diamond, Carlson & Beck, 2005).

> *While Size of Object demonstrates when children can grasp size analogies earlier, Size of Feature stands out as the most difficult condition in this series of studies. Success here requires shifting attention from general appearance to nested object features and this ability develops considerably throughout the preschool years (T. C. Callaghan, 1990; Diamond, Carlson, & Beck, 2016; Shepp, Barrett, & Kolbet, 1987).*

R.1.25
p. 35 – first complete sentence is hard to understand. The working memory points seem unrelated.

Response:
We wanted to point out that searching for relations in the more abstract or complex stimuli requires children to successively focus on different aspects of target and cue (attentional shift) and keep track of the dimensions they have explored (i.e. check for congruence in form, number, orientation). This is where working memory will be beneficial and it is another aspect with crucial developments around age four. We amended the paragraph:

> Second, the reduced accuracy in analogy-based and feature-referenced tasks are both consistent with evidence that younger children tend to focus on surface features rather than relations (Gentner & Hoyos, 2017; Rattermann & Gentner, 1998; Richland et al., 2006), making tasks that require attentional shifting or nested comparison structures particularly demanding (Halford, Wilson, & Phillips, 1998). <u>Shifting attention to different aspects of stimuli sets requires children to keep track of features they have explored.</u> Relational or abstract as opposed to feature-based processing taxes working memory and inhibitory control more strongly (Gick & Holyoak, 1980; Goswami, 1991). This, in turn, points to aspects of cognitive maturation that are likely to foster children's performance in the tasks outlined above, as seen in work on the development of executive functions — particularly the early emergence and developmental trajectories of working memory, inhibitory control, and cognitive flexibility in childhood (De Luca & Leventer, 2010; Diamond, 2013; Reilly, Downer, & Grimm, 2022).

R.1.26
p. 36 – There is some early work on children's production of pictorial symbols showing that children use a wholistic approach and attend to features that will distinguish the referent only later, or if given a prompt that their drawing is difficult to interpret https://doi.org/10.1111/1467-8624.00096

Response:
The publication by Callaghan (1999) is relevant and we are already citing it in the introduction and discussion. In line with suggestions from the other reviewers we would reduce the amount of space dedicated to the production of graphic symbols, but highlight changes from holistic to feature-oriented processing.

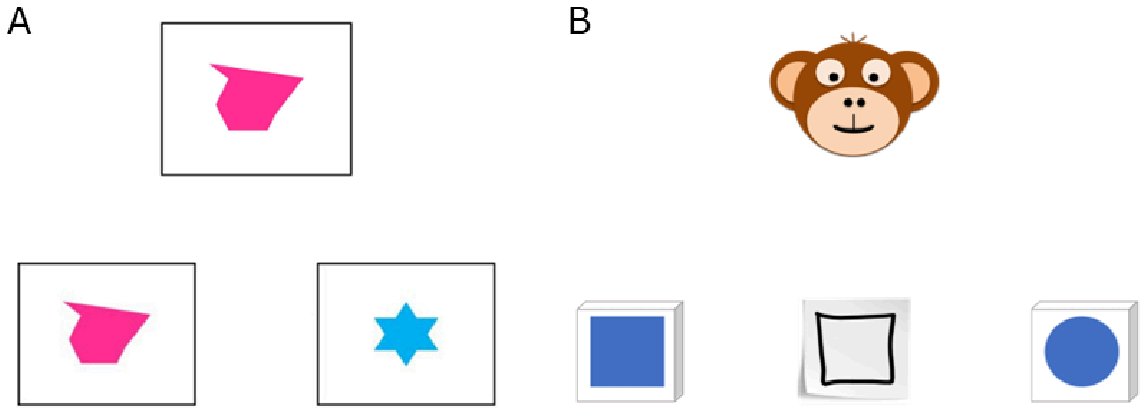
Reviewer 2

Overall evaluation
I commend the authors for the technically well-executed experiments, for the wide variety of relations tested, and for the sound Bayesian analysis methodology. My only problem with the manuscript in its current form is the untested assumption that children interpret the cue–target pairs as symbol–referent pairs. The authors require this assumption in order to argue that the results bear on children's departure from iconicity with age. I elaborate on this below.

R.2.1
*The manuscript claims to be testing "children's ability to interpret graphical symbols" (p. 2). On its own, this sentence is ambiguous between a de dicto reading (children's ability to interpret these visual stimuli, which happen to be graphical symbols) and a de re reading (children's ability to interpret these stimuli *as symbols*). Clearly, they have in mind the second reading: "Our work provides one of the most comprehensive, robust and systematic investigations of young children's ability to find meaning in unfamiliar graphic symbols" (p. 36).*
I suppose the authors assume this interpretation to be likely because of the way the task is set up: a cooperative agent creating drawings on the spot for informative purposes. But there's nothing in the data that supports this interpretation. Indeed, the authors' experiments can be redescribed as match-to-sample tasks just as well. In the absence of a control condition with different results (for instance, one in which there are no drawings involved), there's no way to know whether it matters that children interpret the cues as symbols, or whether they just select the target that exhibits the higher similarity (where similarity is computed along many dimensions) to the cue. In this case, what develops with age is sophistication in how similarity functions are computed. This development might be relevant to how children interpret symbols outside the lab, but it would be a domain-general capacity not specific to symbolic communication.

Response:
We are very thankful for the opportunity to discuss this concern as part of our revision and have rewritten the introduction and presentation of the task to address the problem at hand. We now provide a step by step overview of what is required of children to solve the task in our studies (see page 4).

In a standard **match-to-sample task**, children are presented with two choice items, *match* and *distractor*, and invited to take a closer look. Next, the *sample* item is placed in the middle with the prompt: "Which one of these two goes together with…*experimenter placing the sample* …this card?" (cf. Kroupin & Carey, 2022). Here, the children are directly instructed to identify a match and this is the only inference required. In a **symbolic object-choice-task** children are presented with a hiding game in which a reward is hidden under one of two containers, one serving as the target and one as a distractor. Next, the experimenter directs the child's attention to the task ("And now…look!") and provides a communicative cue. Then the children are provided access to the hiding places and prompted to make a choice ("Do you know where it is?").



A: Typical Identity match-to-sample trial as used in Kroupin & Carey (2022); B: Illustration of a choice trial in symbolic-object-choice task.

In the standard analysis of this setup, children need to understand (1) the pragmatics of the hiding game, (2) that there is a cooperative partner in front of them intending to help, and (3) that the partner is doing so by means of a communicative cue. As a final step, children need to comprehend the cue and connect it to either choice option. In studies on communication, communicative cues may be pointing, markers, replicas, drawings, iconic gesture or switching on a light (Callaghan, 2000; Tomasello, Call, Gluckmann, 1997; Moore, Liebal & Tomasello, 2013; Bohn, Call & Tomasello, 2016). All of these cases may be criticized as merely capitalizing on spatio-temporal contiguity (deixis) or (crossmodal) matching, but they still require communicative inference and pragmatic reasoning.

We are confident that our work provides a crucial contribution to the literature. It exceeds previous work on graphic communication by systematically testing various symbol-referent relationships with robust samples and it goes beyond match-to-sample by testing MTS relations in a communicative context. Please consider also that conditions such as Pars Pro Toto, and the Simple and Complex Form Analogy have no equivalent in the MTS literature. Also we use outline drawings as cues that are more abstract than presenting the cut-out stimulus twice. To cater to the points of the reviewer, we are now devoting more space to discussing the closeness to match-to-sample designs in the introduction by comparing the paradigms (see page 9) at the expense of shortening the discussion of work on graphical communication. Also, please consider this paragraph from the discussion section already present in the first submission:

> *While work on analogical reasoning abilities has highlighted that preschoolers are generally able to identify agreement in graphic displays and specifically involving size, number, and shape also at the age of four (Kroupin & Carey, 2022b, 2022a), this has mostly been established in match-to-sample paradigms where children are conventionally presented with three items and prompted with a phrase such as "which ones go together" directly inducing the cognitive operation tested. In our task, children make matches and comparisons across various dimensions without any verbal scaffolding addressing the match-making process or the dimension of comparison and, hence, showcases abilities for implicit judgements across a wide variety of basic conceptual and visual features. Therefore, our work provides a crucial extension to the literature on analogical reasoning by showing that children can actually put their developing skills to work in the practical context of a versatile communicative coordination game requiring both relational reasoning and pragmatic inference.*

R.2.2

Do the authors expect different results if children were given the same stimuli and asked instead to select the one that is more similar? If yes, then the authors have to argue that there is no other way to account for the results than by positing that the implicit nature of the task reveals interpretation of the cue as a symbol and of the banana container as its referent. If not, what would be the different results and why? [Counterarguing that there is extensive evidence that children interpret drawings as symbols would only go so far, since there are no actual drawings in the study—the cues are, in fact, representations of drawings, which children must infer from the schematic animations on the tablet: by this age, they know very well that there is no actual drawing monkey in front of them.] Let me state explicitly that this does not draw anything away from the findings themselves, which are interesting regardless of how they are framed. But, unless I am missing something, it unclear to what extent the findings can inform symbol–referent relations in cognition.

Response:

We appreciate that the reviewer acknowledges that this is an issue regarding the framing of this work and that it does not draw anything away from the findings. As we are now explicitly outlining in the abstract and introduction, our task requires both analogical reasoning (match-to-sample) and pragmatic/communicative inferences. As such our work connects both strands of literature.

Much like reviewer 2, we were aware of the tension and very interested in addressing this intersection directly. In an ongoing project, we are currently comparing the influence of communicative framing (symbolic object-choice-task) and non-communicative framing (standard match-to-sample) on children's performance with graphic stimuli. Preliminary results suggest that preschoolers perform considerably better when the same classic match-to-sample stimuli are used in standard framing as opposed to a communicative framing. Our preferred interpretation is that the communicative framing is more demanding as it requires pragmatic inference (keeping track of the context, not being told what to do). However, as of now these results are still pending.

In addition to having reworked how the studies are presented in the introduction, we now also address this in the discussion as a goal for future investigations.

> *The research described here has limitations that future research needs to address. The combinations of outline drawings and geometric shapes we used here could also be presented in a standard match-to-sample procedure in which children are solely asked to find a match between cue and target. When directly instructed to align stimuli and without the need for considering context, children may succeed earlier. To connect the work presented here with the literature on analogical reasoning, future work should present children with the same set of stimuli in the context of both paradigms.*

R.2.3

The authors make an observation in passing that is relevant for this question: "Success even prior to the fourth birthday in these conditions is remarkable as it requires children to identify and compare relations (Absolute Position – figure/ground; Relative Position – figure/figure) which is demanding even later throughout the preschool years in other contexts (Shivaram, Shao, Simms, Hespos, & Gentner, 2023)" (p. 32). If struggle with relational matching children in nonsymbolic contexts but find it easy in symbolic contexts (assuming identical tasks), the authors' argument would be much strengthened. However, the CogSci proceedings paper the authors cite reports successes, not struggles, on relational match-to-sample tasks in 4½-year-olds.

Response:

Yes, Shivaram and colleagues (2023) report success in 4½-year-olds, but specifically against the background of failure of the same age-group with highly similar stimuli in Christie and Gentner (2014) and Kroupin and Carey (2022). We add all three references but keep the phrasing because RMTS is indeed demanding for preschoolers and results are mixed. As for the comparison of our findings and the RMTS literature, the stimuli in *Absolute Position* and *Relative Position* are relational by drawing on Gestalt principles (figure-ground; proximity) but are still more simple than classic RMTS tasks or the stimuli in study 3. This is what we wanted to draw attention to as this is again a valuable contribution for researchers interested in graphic communication, map-reading, Gestalt-Principles but also RMTS.

R.2.4

I can think of several ways to address this: 1) If I missed something, point it out (and modify the manuscript such that other readers don't fall into the same trap). 2) Find evidence in the current data or arguments from the literature that children do interpret the visual cues as symbols (e.g., because they perform better on these tasks than on more standard match-to-sample ones). 3) Adopt a more moderate position, and claim that the experiments test relations of various sorts systematically, then spell out the potential implications of the findings for symbolic competence, iconicity development, and so on in the General Discussion. This option would require de-emphasizing the literature review on symbols in the Introduction.

Response:

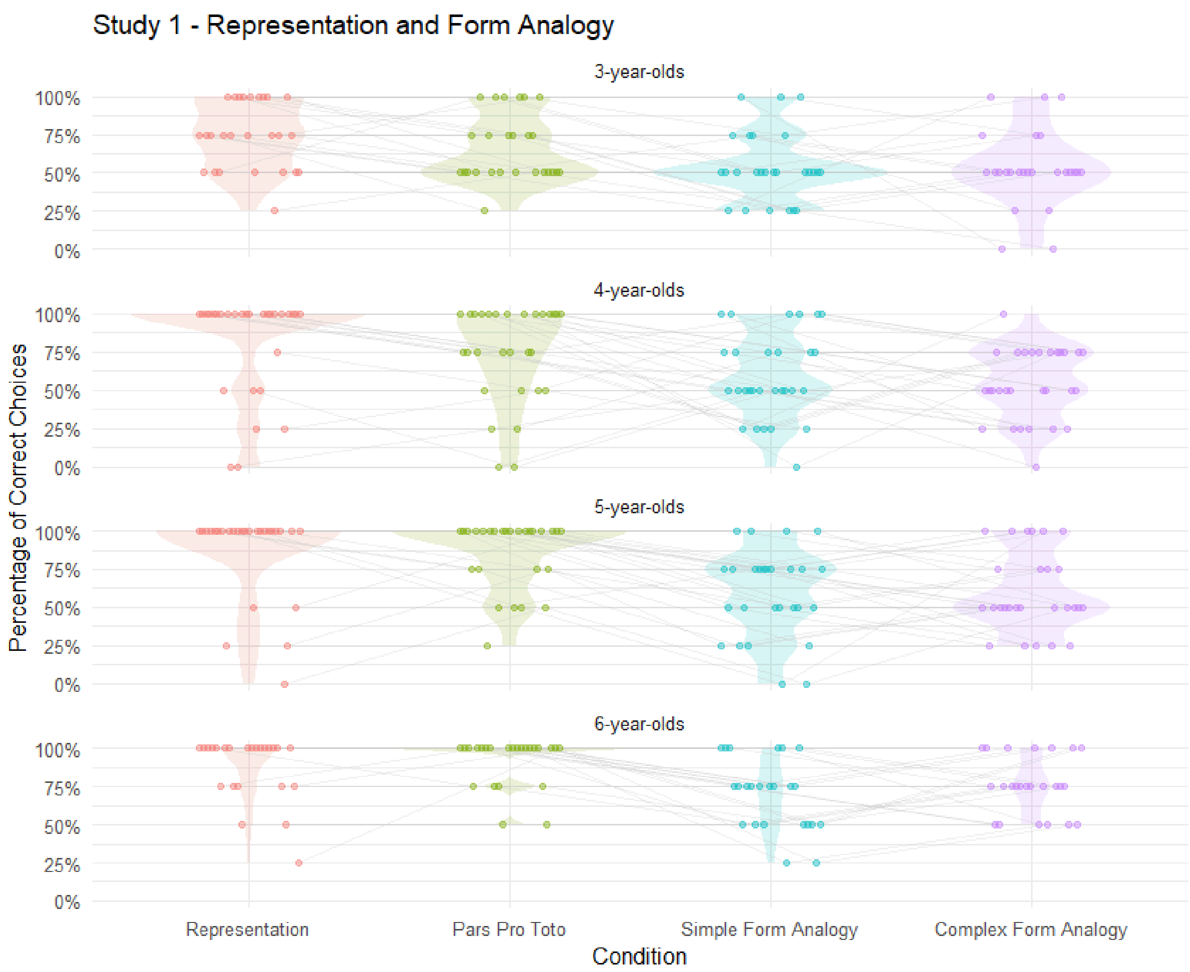
Any object or action can be a symbol or utterance when used with communicative intent. In the context of the symbolic object-choice-task we use, it would be misguiding to not use the term symbol-referent-relationships when discussing the many ways that motivate a child to see how cue and target are matching, and there is hardly a trap here: the introduction - in its original form and more so now - specifically refers to match-to-sample tasks several times, we frequently highlight that the relationships we test are based in analogies and the discussion also devotes several paragraphs referring to analogical reasoning and match-to-sample tasks, including this paragraph explicitly addressing the point under discussion.

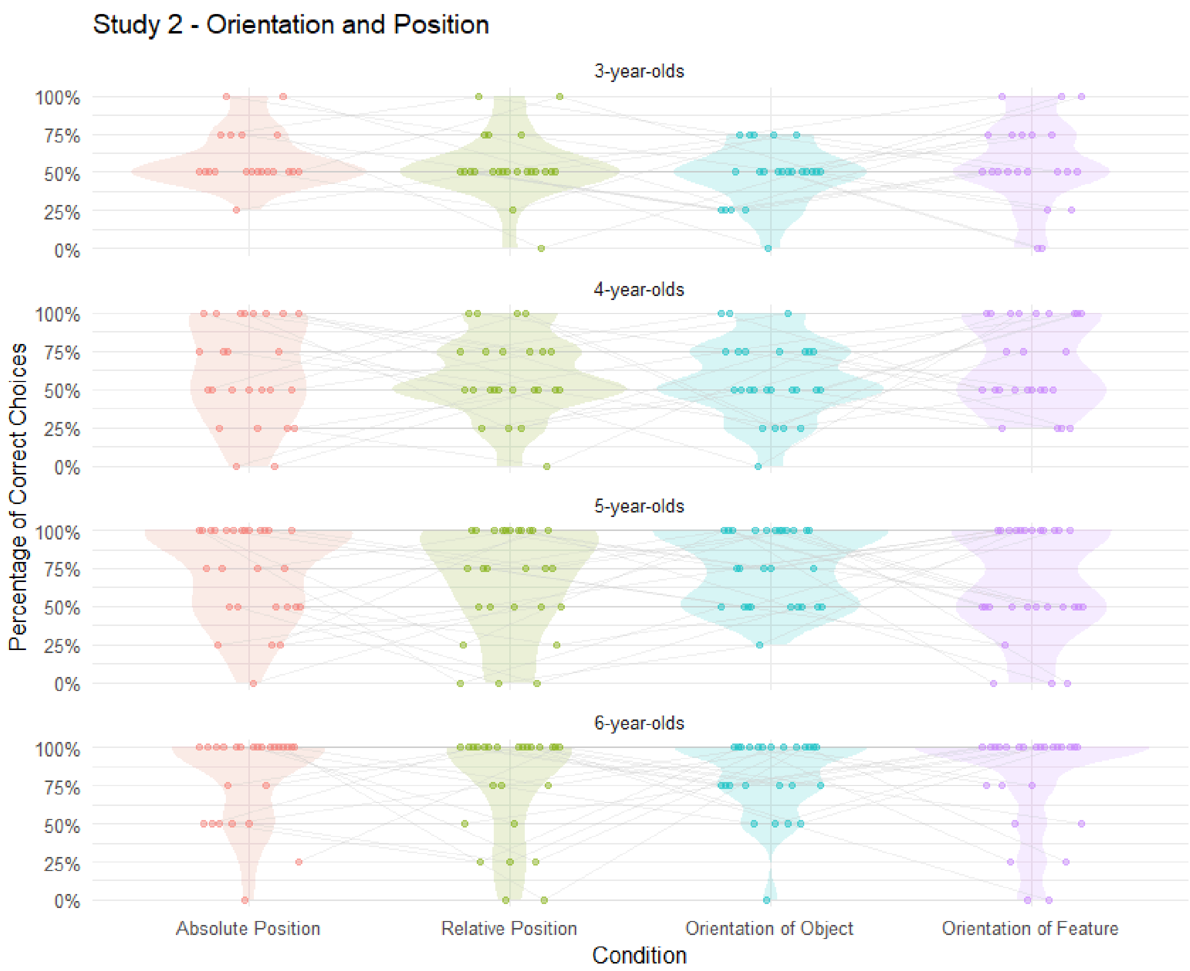
R.2.5 Data visualization

The results figures for all three experiments are currently plotted between-subjects, as it were (since they plot the results in each condition separately). But since condition was within-subjects, it would have been great to have plots illustrating performance in each condition by age group (one plot/facet for 3-year-olds, one plot/facet for 4-year-olds, etc.). Each of these subplots would show the performance of individual participants across conditions (from the most iconic to the most abstract), connected with lines. This would show at a glance how the iconic–abstract gradient affects performance within single subjects. Whether supplementary analyses are required is hard to tell (perhaps the data is clear enough that visual inspection alone will do). I am sure the authors can make a sensible decision about this.

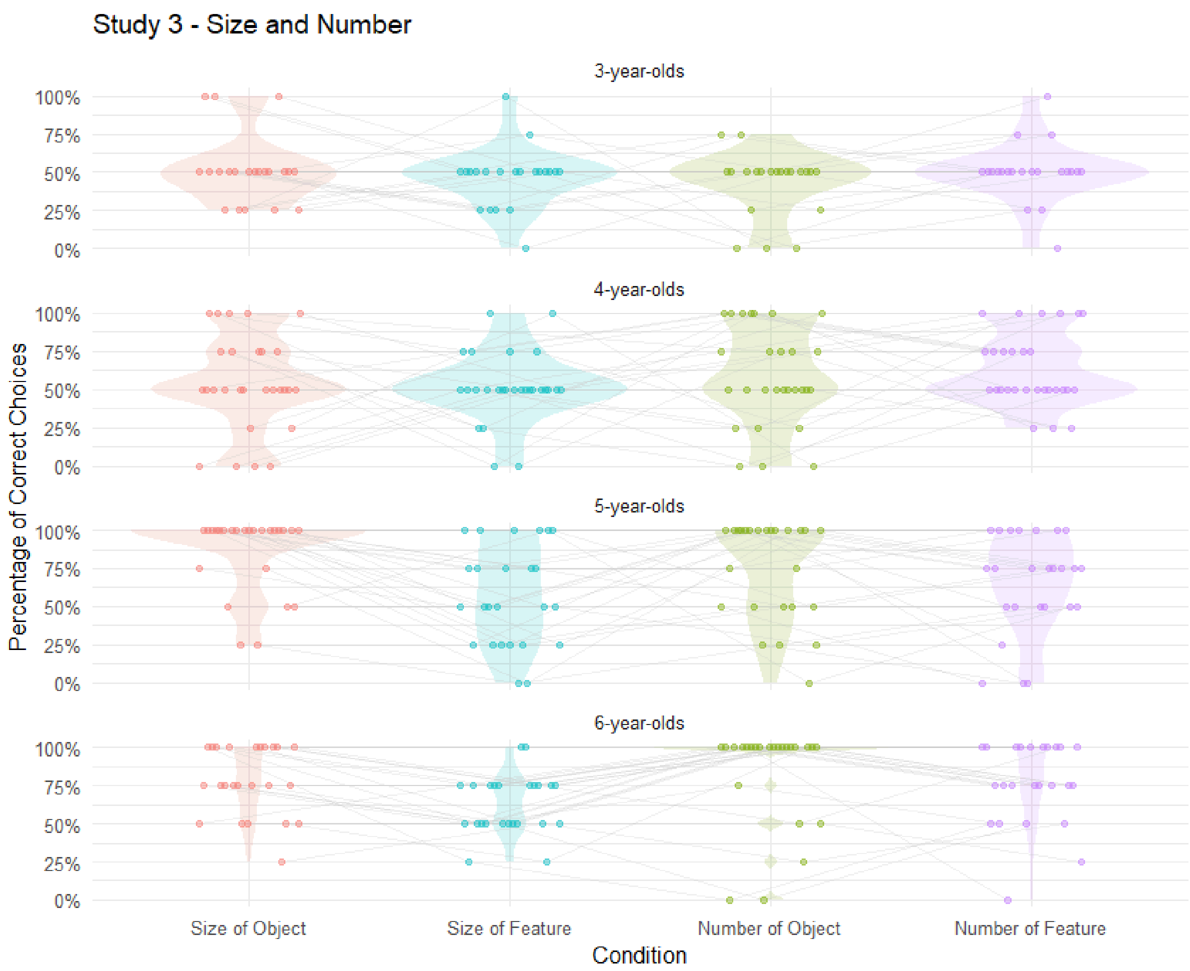
Response:

Below, please find plots faceted by year of age. In the paper we avoid binning participants by year of age as this would entail losing information. Our goal was specifically to sample participants continuously throughout the preschool years and draw smooth developmental curves for the respective conditions. We then decided to facet by condition (cf. figure 2,3,4) as there was too much overlap between conditions. To us, no clear picture emerges from the three novel plots below, which is why we would opt against including them in the manuscript or supplements.





Study 2 - Orientation and Position
3-year-olds
4-year-olds
5-year-olds
6-year-olds
100%
75%
50%
25%
0%
Percentage of Correct Choices
Absolute Position
Relative Position
Orientation of Object
Orientation of Feature
Condition



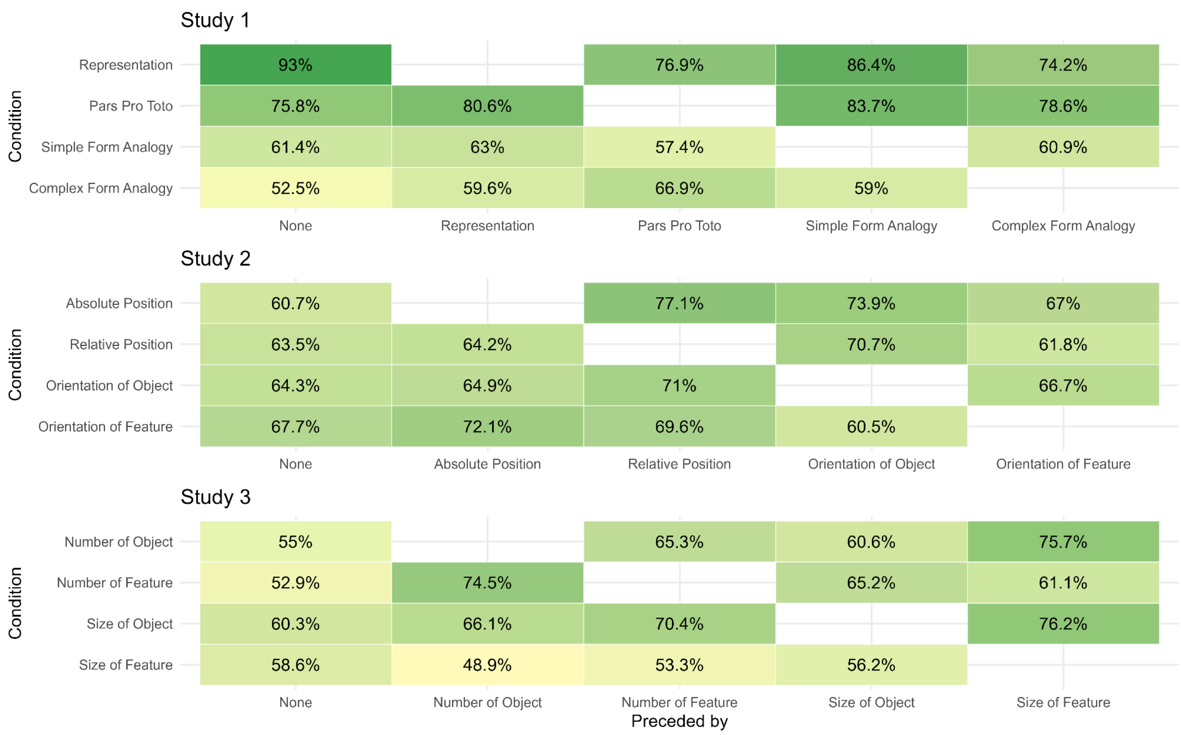
R.2.6 Supplementary analysis

The task of Study 1 has one pragmatically curious aspect, which may affect performance in interesting ways. Here's what I mean: In Study 1, the targets are repeated across conditions (e.g., circle versus square), which makes the monkey's decision to opt for different drawings (despite communicating the same content) somewhat strange. Could this contribute to younger children's struggle with the task? One potential way to address this would be to check whether there are order effects and whether they interact with children's performance from earlier to later trials: For instance, are the children who saw the representation condition first more likely to err on pars pro toto, simple-form analogy, and complex-form analogy later on? (To illustrate: If the monkey is clearly able to draw a circle, why does she turn to a scribble? This could give rise to a mutual exclusivity inference and thus to selecting the 'wrong' answer.)

Response:

To an extent, the reviewer's comment applies to the experimental situation in a large amount of studies on communicative development: Isn't it odd to solely point silently instead of telling the child where the toy is hidden? Why is the experimenter using iconic gestures instead of telling the child how to solve the riddle box? In our case, the reviewer highlights that the repeated use of similar stimuli may be particularly problematic. Generally, as there is always the possibility to provide a clean drawing of the correct target, the monkey is violating the Gricean Maxime of quality (be as informative as possible) and/or manner (be clear and orderly; avoid obscurity) in all conditions except for *representation*. We think that this is necessary for testing specific symbol-referent relationships in isolation but in a within-subjects design and not irritating in the context of a playful picture-book-style developmental study.

To address order effects, we have aggregated data from all studies and provide the percentages of correct choices for each condition (y-axis) separately depending on the preceding condition (x-axis). For example in study 1, *Representation* has the highest performance when presented in the first block. *Pars Pro Toto* has a slightly higher proportion of correct choices when preceded by *Representation*, which is reasonable as it is just the reduced version. At the same time *Pars Pro Toto* has its highest values when preceded by *Simple Form Analogy*, for which I can provide no good explanation. *Simple Form Analogy* yields almost identical results regardless of its position in the counterbalancing and *Complex Form Analogy* is hardest in first position and benefits more from being preceded by *Pars Pro Toto* than *Representation*. So at least it is not detrimental to children's performance to go from a condition with clear "utterances" to another one violating Grice's maximes. All in all, the picture that emerges is not clear and switching effects appear moderate overall.



Please consider, the overall developmental trajectories we report as our main finding may not be influenced by switching costs as variance due to trial order has been controlled for and the order of conditions is counterbalanced. It is likely that EF skills may impact switch costs at an individual level, which is a good justification for including random slopes for individuals as we did.

R.2.7

In Study 3, there are no target repetitions but the symbols are shared between condition pairs (size of object – size of feature and number of object – number of features). Are children more likely to make a mistake when a symbol is repeated?

Response:

In both cases, children are actually performing better when the conditions are preceded by each other than when occurring in the initial position (see plot above). When doing two size or number blocks successively, this helps children as it may reduce the cost of switching across tasks. Children do not appear irritated by symbols being shared across conditions.

R.2.8

Finally, also in Study 3, the crossing of the square and the circle also becomes pragmatically odd, as children progress through the task: If the monkey can draw a circle (on trials with squares) why does she not draw one on circle trials? Evidence of pragmatic inferences could, I think, also bolster the authors' claim that children interpret the drawing as symbols of the referent locations (at the very least, it would show that children interpret the drawings communicatively).

Response:

Reviewer 2 suggests that the crossing of circles and squares: a big circle referring to a big square in contrast to a small square, and then vice versa (cf. the first two columns in the matrices below) may be particularly confusing or demanding for children in consecutive trials. We are very thankful for this thorough reading of the manuscript and for taking the time to actually picture what participating in this study may be like for children. However, we see no issues here with regard to the study methods or the interpretation of results.

First of all, this monkey is not a good Gricean communicator and this is a compromise we were willing to make when designing the study in order to test relatively abstract symbol-referent relationships. We now acknowledge this directly in the introduction when discussing the setup (page 6).

Generally, counterbalancing the order of conditions should ensure that all conditions may be equally affected by potential effects of order. We have opted to cross circles and squares across trials in study 3 as they are the most simple and canonic shapes - hopefully reducing processing demands - and for also being the most distinct: matches between cue and target may only be made on the basis of size but are completely implausible with regard to shape.

If there would be problems with regard to trial order, either trial version should be equally affected as the order of trials and conditions were counterbalanced. The problematic transition under discussion has a 1 in 4 probability of occurring. Furthermore, children would have to track stimuli across trials to be irritated by this. If so, they may for example get frustrated and stop earlier as the monkey appears to be an irrational communicator. However, drop-out rates are negligible (study 3: one exclusion for not completing at least eight out of 16 test trials; three exclusions due to fussiness) and results are controlled for trial order.

We think it likely that children actually enjoy such communicative games when containing puzzles. Much like when playing charades where not being able to talk is particularly exciting, not using the most obvious way of graphic reference in the context of our hiding game may be interesting too. Children are also very familiar with not directly being provided with solutions in hiding games, but rather receiving playfully suggestive cues.

R.2.9 A potential reason for the difficulty of some conditions
I was also wondering whether it matters that children received each target pair only once in each condition, paired with a single cue. So children lacked access to the contrast between the two alternatives the authors designed. I guess the authors expect that performance would improve if children witness the monkey choosing between the two options? (It is clear to me, for instance, which cue goes with each target in the complex-form analogy condition of Study 1 because I can clearly see the contrast between the round and the spiky cue, but I am not sure I would have converged on the 'correct' interpretation without it.) It seems that this is worth mentioning in the General Discussion.

Response:
We agree that this makes the tasks more difficult and it was an intentional choice to present children with a novel cue-target pairing in each trial (and in the absence of feedback etc.). We think this is ideal for investigating which symbol-referent-relationships children can decode spontaneously. Providing the contrasts in question would have made it significantly easier but also changed the task considerably. But consider that children saw four trials per condition in a blocked fashion. For example, in the *size condition* this would entail seeing four cases of the monkey drawing a novel cue that was either large or small to refer to targets differing solely in size. Instead of adding to the discussion that performance could have been increased by presenting pairs of cues that the monkey could choose from (or providing feedback, training match-to-sample first, providing more verbal scaffolding, etc), we would take the liberty of adding to this point to the discussion when highlighting why our paradigm is a conservative test of children's abilities (page 31).

Reviewer 3

Overall evaluation

The research questions examined are theoretically relevant and the findings will be of great interest to developmental psychologists. The presentation of the manuscript could be greatly improved to more clearly motivate the research questions and situate the findings in the literature, as I will detail below.

R.3.1

1. I understand why the authors use the term "graphical symbols" as the paradigm involves the generation of the symbols by the monkey. I think it would be helpful to more clearly outline the use of the term early in the manuscript to help the reader appreciate what is meant by graphical.

Response:

We have reworked the introduction and added a paragraph providing a detailed discussion of symbolic-object-choice tasks together with the inferences required by children to solve these tasks. Here we list various types of cues and introduce the term "graphic symbols" (see page 4).

R.3.2

2. In a number of places in the manuscript, including the keywords, the term pragmatics is used. I am not sure in what sense this term is being used in this context and it would be helpful to clarify early on in the manuscript.

Response:

Thanks! The introduction now contains the aforementioned paragraph providing a set-by-step analysis of the inferences children need to make when solving the symbolic object-choice tasks we used in our tasks. Here we specifically introduce *pragmatic inference* as a way of referring to the step where children have to integrate the demands of the search game, the perceived helpful intent of their interlocutor, and the perceived similarity between the interlocutors drawing and one of the hiding places, to arrive at the conclusion that the drawing is a cue referring to a target (see page 4). Hence, pragmatics here does not narrowly entail linguistic concepts such as implicature (Grice) or illocutionary force (Austin) which may also have been plausible associations for readers with the term *pragmatics*.

R.3.3

Much of the literature review focuses on children's interpretation of drawing, which is of course, highly relevant to the research questions. The discussion of the role of shape could, however, be more concise (for example, I am not sure why there is discussion of shape and sound-symbolism), with more space devoted to discussing the mechanism that underly children's interpretations of drawing. For example, I expected more consideration of work by

Gentner and colleagues around comparison and analogical reasoning, given the paradigm used invites children to engage in comparison reasoning to identify the correct target.

Response:
Prior to revision our goal was to provide an in-depth overview of children's production and comprehension of the symbol-referent-relationships tested in the three studies, but we agree with the editor and reviewers that this was not ideal. We have restructured and shortened the introduction focusing on describing object-choice tasks and contrasting it with match-to-sample paradigms. References to children's production of drawing have been excluded for the most part.

R.3.4
I appreciate that the studies were pre-registered. Were the sample sizes and, in particular, the goal of testing 2 children per month between 3 and 7 years of age, based on power analyses? If yes, it would be helpful to note this in the manuscript.

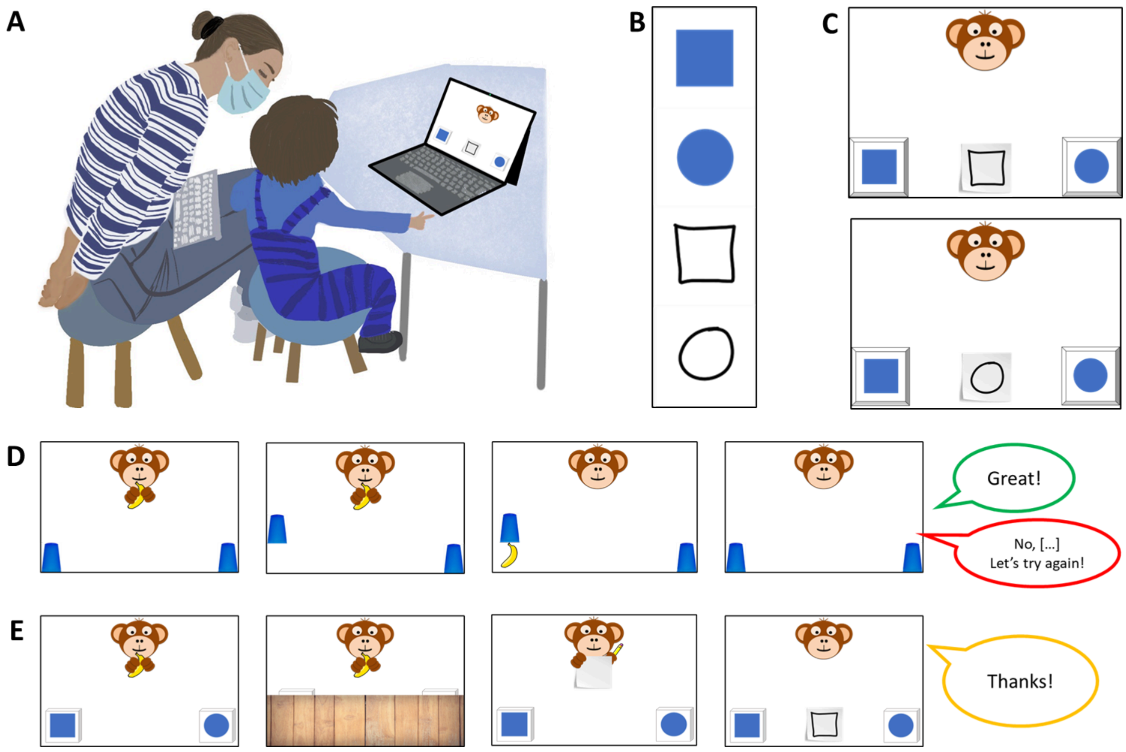
Response:
In the preregistration we stated:

> *Starting at 36 months, we will test two children per month over a range of four years (48 months) for a sample of 96 children. While using age as a continuous predictor is less common in developmental studies, this design yields a sample of at least 24 children per year of age which is in line with conventions in the field.*

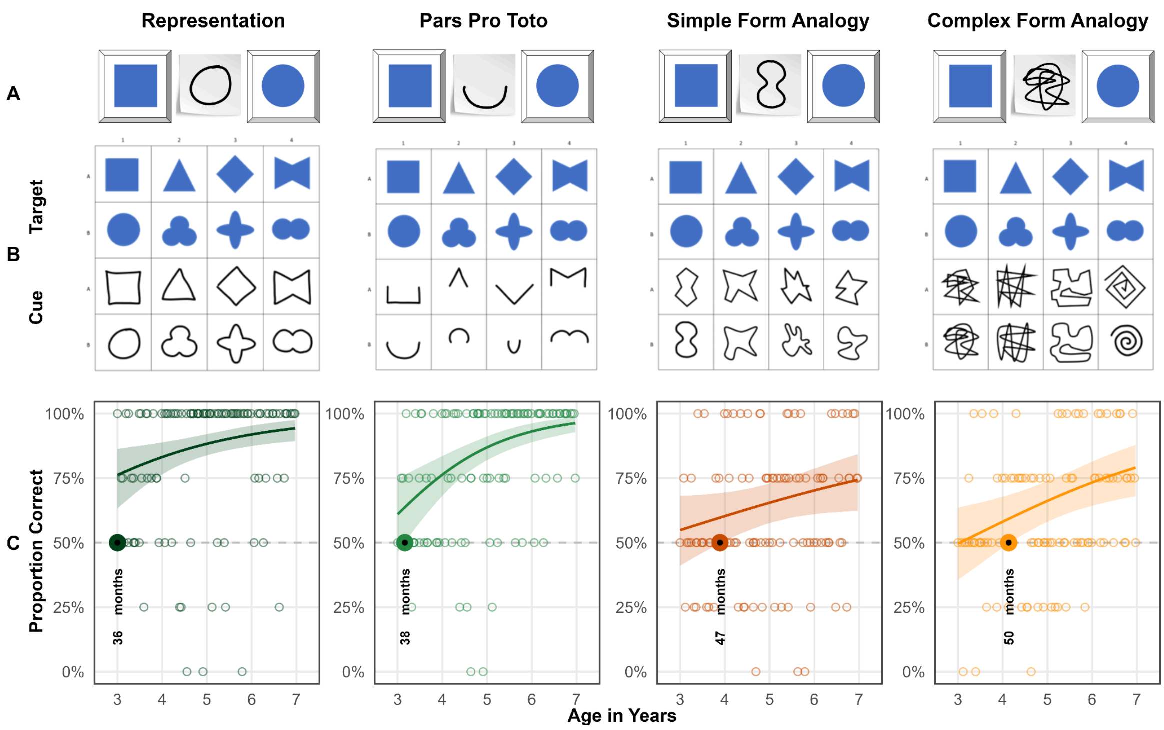
Our main aim was to collect a balanced continuous sample to determine the ages at which children succeed with the respective symbol-referent conditions. At the time of preregistration we did not have estimates for effect sizes in this paradigm and analytic approach, such that we opted for default priors and a sample aligning with conventions in the field.

R.3.5
I appreciate the figures depicting the results, however, it would greatly help the reader if there was a figure earlier in the methods depicting the different types of representations presented. It really helps understand what the child is seeing and what the researchers are testing.

Response:
The general methods section contains a figure (see below) that illustrates (A) the setup with which children were tested, (B) an example of cue and target stimuli of a specific trial and (C) how they were arranged to create the test slides. Finally, we show (D) how these would be inserted into a test trial.



We have opted to present all of the stimuli together with the results as this is particularly helpful when evaluating them. To guide the reader, we have added a reference to figures 2, 3, and 4 respectively when discussing stimulus design in the general methods section.



R.3.6

On p. 20, the authors indicate that "children perform above chance… as early as 36 months" . Later they state "children succeed only at 38 months in Pars Pro Toto". Given that the chance-level analyses are based on ages binned by years and only 2 children are tested exactly at each months, these statements are not quite accurate and should be revised. This comment applies throughout.

Related to the comment above, it is not quite clear which analyses the authors are using to determine the age at which children succeed—can this be made clearer? I am assuming it is the chance-level t-tests, as age does not interact in the models in Expt 1, except for the Pars pro toto condition. It is possible that I am not clear on how the Bayes models are being conducted.

Response:

The children are only binned according to their age in years for the additional conventional analyses in the supplements. In the supplements we provide conventional t-tests for the convenience of readers that are less familiar with the main analytic approach and may want to compare results. This is mentioned in the last paragraph of the methods section.

Our main analyses are Bayesian GLMMs that allow us to continuously model the likelihood of children making correct choices across the entire age-range from three to seven years of age. These analyses are not restricted by bins according to years of age. In the methods section we then also describe how we determine the age of success:

> *To answer the main research question of when children as a group systematically make correct choices in any of the conditions outlined below, we followed the approach of House and colleagues (House et al., 2013) and plotted the model estimates graphically as a smoothed curve surrounded by the respective confidence intervals. We fitted models to predict the developmental trajectory (with 95% CrI) of group level performance drawn from values of the posterior predicted distribution via the function fitted. These trajectories and CrIs were plotted by age. The criterion for settling when children performed above chance was the point at which the lower bound of the 95% CrI for a particular trajectory did no longer overlap with a mid-line demarcating the 50% chance level.*

Hence, we are able to provide very robust estimates of the age in months when group level performance exceeds chance level and this is central to the three studies presented. We made amendments to the manuscript to better guide the reader:

- We added a paragraph break when introducing how we determined the age of success. The paragraph opens with "To answer our main research question of when children perform above chance [...]"
- When referring to the additional exploratory t-tests in the supplements, we specifically highlight that these are only used to determine whether an age-bin is

above chance and are only added in the supplements for readers that might want to compare our main analyses with conventional t-tests.
- We now directly highlight the “age of success” when referring to the result figures of study 1, 2, and 3.

R.3.7

The Discussion section provides a detailed review of the results. I think this could be condensed, as it repeats what is noted earlier, and more attention could be devoted to the mechanisms that support children’s reasoning in these studies (as noted earlier, I would suggest comparison and analogical reasoning be discussed in more detail).

Response:

We followed the suggestions of reviewer 3 to reduce redundancy by shortening the interim summaries following each study, as well as the final summary. We have introduced several new references and work by Gentner and others. We found a paper by Cooperrider and Goldin‐Meadow (2017) particularly fitting that discusses relations between gesture and analogical reasoning as it also connects the literature on analogical reasoning with a communicative phenomenon. The authors distinguish gestures capturing attributes with others capturing relations and refer to the latter as analogical gestures. This ties in with the object/feature distinction we use (i.e. size of object / size of feature) as well as the difference between. Our work generally ties in with basic tenets of Gentner and Hoyos (2017) such as “[...] the more different the two comparands, [...] the less likely children are to spontaneously compare [...] and align them.”

Cooperrider, K., & Goldin‐Meadow, S. (2017). When gesture becomes analogy. *Topics in Cognitive Science*, *9*(3), 719-737.

Gentner, D., & Hoyos, C. (2017). Analogy and abstraction. *Topics in cognitive science*, *9*(3), 672-693.

Winter, B., Woodin, G., & Perlman, M. (2026). Defining iconicity for the cognitive sciences. In O. Fischer, K. Akita, & P. Perniss (Eds.), *The Oxford handbook of iconicity in language* (pp. 138–150). Oxford University Press.

R.3.8

The authors propose that the results signal a “qualitative shift” in reasoning. I would suggest that this conclusion needs to be better supported than it currently is. It is very difficult, based on the data provided, to distinguish a qualitative shift from quantitative growth. The figures depicting the developmental growth do not suggest abrupt or discontinuous changes.

Response:

We agree that our visualisations and results are strictly speaking not sufficient for separating a quantitative from a deep qualitative shift in the underlying cognitive processes. However, it is remarkable that across three studies with different samples and across a versatile set of conditions, we find evidence for success around the fourth birthday. We have replaced all instances of the phrase “qualitative shift” with “major shift” in order to draw the reader's attention to these converging results without specific implications as to the underlying cognitive developments.